

Unified Evaluation of Reactive and Map-Based Autonomous Navigation Strategies with TTC-Based Safety Supervision in ROS2

Santiago Estrada[✉], Martín Melguizo, Jhoan Chacón, and David Rozo-Osorio[✉]

Abstract—This paper presents a unified simulation-based evaluation framework for autonomous navigation of differential-drive robots integrating both reactive and map-based navigation strategies within a ROS2 and Gazebo architecture. The evaluated methods include wall-following, distance-based Follow-the-Gap (FTG), TTC-based FTG, Pure Pursuit, and Stanley controllers operating under a common Time-to-Collision (TTC)-based Automatic Emergency Braking System (AEBS) supervisory layer. Reactive approaches generate control actions directly from LiDAR observations, whereas map-based methods rely on an ARA*-generated reference trajectory. The comparison focuses on navigation smoothness, tracking behavior, traversal efficiency, and safety-related interventions under identical obstacle-constrained environments. The results show that Pure Pursuit achieves smoother path tracking and fewer safety interventions than Stanley, while the TTC-based FTG method exhibits the most consistent reactive behavior by combining anticipatory obstacle avoidance with zero AEBS activations. The study highlights the influence of corrective aggressiveness and TTC-aware decision mechanisms on overall navigation performance and provides a compact, reproducible framework for comparative evaluation of autonomous navigation strategies.

Index Terms—Autonomous navigation, mobile robotics, reactive navigation, path tracking, time-to-collision

I. INTRODUCTION

AUTONOMOUS navigation is a fundamental problem in intelligent robotics, requiring the integration of perception, planning, and control to enable safe and efficient motion in complex environments. In academic and research settings, simulation-based mobile robotics platforms provide a controlled and reproducible framework for evaluating autonomous navigation strategies under identical operating conditions while avoiding the cost, safety risks, and hardware limitations associated with physical experimentation. In particular, differential-drive robots implemented in ROS2 and Gazebo environments enable rapid development and comparative assessment of navigation algorithms, making them suitable for studying the trade-offs among tracking accuracy, obstacle avoidance, motion smoothness, and traversal efficiency.

Autonomous navigation strategies for wheeled mobile robots can be broadly categorized into reactive and map-based approaches. Reactive methods generate control actions directly from onboard sensor feedback, enabling fast local responses in partially known or dynamic environments. Classical approaches such as the Dynamic Window Approach formulate obstacle avoidance in the velocity space while respecting dynamic constraints [1], whereas behavior-based architectures combine goal tracking and obstacle avoidance through parallel reactive modules [2]. Follow-the-Gap (FTG) methods instead identify collision-free sectors in LiDAR data and steer the

robot toward the most traversable region, providing computationally lightweight obstacle avoidance suitable for real-time applications [3]. Other reactive strategies rely on wall-following and boundary-tracking formulations to maintain stable motion relative to surrounding obstacles [4], [5]. Recent extensions have further incorporated cooperative and dynamic planning capabilities into reactive navigation frameworks [6].

In contrast, map-based navigation methods rely on a pre-computed representation of the environment and a reference trajectory that must be accurately tracked through dedicated control laws. Geometric path-tracking controllers such as Pure Pursuit and Stanley are widely used due to their conceptual simplicity and effectiveness in structured environments [7], [8]. Pure Pursuit computes curvature commands from a look-ahead target point along the reference trajectory, promoting smooth path convergence with relatively low computational cost [9]. Stanley control instead combines heading and cross-track error feedback through a nonlinear steering law capable of reducing trajectory deviation under varying operating conditions [8]. In parallel, search-based global planners such as Anytime Repairing A* (ARA*) provide computationally efficient path generation with bounded sub-optimality guarantees, making them suitable for autonomous robotic navigation pipelines [10].

Despite the wide variety of autonomous navigation methods available in the literature, comparative evaluations between reactive and map-based strategies under a unified experimental framework remain relatively limited, particularly when considering safety-supervisory mechanisms. In intelligent robotic systems, Automatic Emergency Braking Systems (AEBS) based on Time-to-Collision (TTC) estimation provide an additional decision layer capable of overriding nominal commands under imminent collision risk [11]. However, the interaction between navigation strategies and TTC-based safety supervision can significantly influence motion smoothness, efficiency, and overall navigation behavior.

This paper presents a unified simulation-based evaluation framework for autonomous navigation of a differential-drive robot integrating both reactive and map-based navigation strategies within a common ROS2 and Gazebo architecture. The study compares representative reactive methods, including wall-following and Follow-the-Gap variants, against map-based path-tracking controllers based on Pure Pursuit and Stanley formulations. Additionally, a TTC-based AEBS supervisory layer is incorporated to analyze safety-performance trade-offs under identical experimental conditions. The main contribution of this work is a reproducible comparative framework that enables systematic evaluation of autonomous navigation strategies while explicitly characterizing the relation-

ship among tracking behavior, collision avoidance, navigation smoothness, and safety intervention frequency.

The remainder of this paper is organized as follows. Section II describes the simulation framework, robot platform, TTC-based AEBS, and navigation architecture, alongside the evaluated reactive and map-based navigation methods. Section III presents the experimental setup used for evaluation. Section IV reports the experimental results and comparative analysis. Finally, Section V summarizes the main findings and outlines future work.

II. METHODOLOGY

A. Simulation Framework

The proposed navigation framework was implemented in ROS2 Jazzy and evaluated in Gazebo Sim (Harmonic) under controlled and reproducible simulation conditions. The robotic platform was modeled using URDF/Xacro and integrated with the simulator through the `ros_gz_bridge`. The simulated environments consist of static obstacle-constrained scenarios represented through occupancy maps and collision meshes, allowing consistent comparison of navigation strategies without the influence of dynamic obstacles or stochastic interactions.

The mobile platform follows a differential-drive configuration with two independently actuated wheels under standard nonholonomic constraints. The robot is equipped with a 360° LiDAR sensor providing 720 range measurements per scan and an IMU for orientation estimation. These sensors provide the perception information required by both reactive and map-based navigation methods.

In the proposed architecture, reactive methods operate directly from instantaneous sensor observations, whereas map-based methods rely on a precomputed reference trajectory generated from an occupancy-grid representation of the environment. The resulting framework enables unified evaluation of different navigation paradigms under a common robotic and simulation infrastructure.

B. TTC-Based Automatic Emergency Braking System

An Automatic Emergency Braking System (AEBS) was incorporated as an independent supervisory safety layer operating alongside all evaluated navigation methods. The system continuously monitors commanded robot velocities and LiDAR measurements to estimate collision risk and temporarily override nominal navigation commands when necessary.

Collision risk is evaluated using the Time-to-Collision (TTC) metric, defined as the predicted time remaining before impact assuming constant relative velocity. For a LiDAR beam at distance r_i and relative radial velocity $\dot{r}_i$, TTC is computed as

$$TTC_i = \frac{r_i}{\dot{r}_i}. \quad (1)$$

In practice, the relative radial velocity is obtained from the projection of the commanded longitudinal velocity onto the beam direction, yielding

$$TTC_i = \frac{r_i}{v_x \cos \theta_i}, \quad (2)$$

where v_x is the commanded forward velocity and θ_i is the LiDAR beam angle. This projection-based formulation is conceptually related to projected Time-to-Collision approaches used in intelligent transportation safety systems [11].

At each control cycle, the minimum TTC and minimum obstacle distance within the robot heading direction are evaluated against predefined safety thresholds. If a critical condition is detected, the AEBS overrides the commanded forward velocity and enforces an emergency stop while preserving rotational motion for evasive maneuvers. Since the AEBS operates independently of the navigation controller, it provides a unified safety-supervision mechanism for both reactive and map-based navigation strategies.

In the implemented AEBS, emergency braking was triggered when the minimum TTC within a $\pm 20^\circ$ frontal sector fell below $TTC_{\min} = 0.5$ s or when the minimum obstacle distance dropped below $d_{\min} = 0.6$ m.

C. Map-Based Navigation

The map-based navigation pipeline relies on a global path generated from an occupancy-grid representation of the environment. Path planning is performed using the Anytime Repairing A* (ARA*) algorithm, which extends classical A* search through an inflated heuristic formulation capable of producing bounded sub-optimal solutions with reduced computational cost [10].

The planner evaluation function is defined as

$$f(s) = g(s) + \epsilon h(s), \quad (3)$$

where $g(s)$ is the accumulated path cost, $h(s)$ is the Euclidean heuristic distance to the goal, and $\epsilon \geq 1$ is the heuristic inflation factor controlling the trade-off between optimality and search efficiency.

The resulting reference trajectory is tracked using two geometric path-following controllers: Pure Pursuit and Stanley. Pure Pursuit computes curvature commands toward a look-ahead point located along the reference trajectory, promoting smooth trajectory tracking with relatively low computational complexity [9]. In contrast, the Stanley controller combines heading and cross-track error feedback through a nonlinear steering law that promotes tighter path convergence [7], [8]. Both controllers adapt linear velocity according to tracking conditions to improve stability and reduce oscillatory behavior in high-curvature regions.

D. Reactive Navigation Methods

Three reactive navigation strategies were implemented and evaluated: wall-following, distance-based Follow-the-Gap (FTG), and TTC-based Follow-the-Gap. Unlike map-based navigation, these approaches generate control actions directly from instantaneous LiDAR observations without relying on a global trajectory representation.

The wall-following strategy estimates the orientation and distance of a nearby boundary using right-side LiDAR measurements and regulates the robot motion through proportional-derivative feedback inspired by geometric wall-tracking formulations [4]. Linear velocity is simultaneously

modulated according to tracking error magnitude to improve corner negotiation and stability.

The Follow-the-Gap methods instead evaluate obstacle-free angular sectors within the LiDAR scan and steer the robot toward the most traversable region [3]. The distance-based implementation selects gaps according to geometric clearance and free-space width, whereas the TTC-based formulation evaluates admissible sectors using collision-time estimates rather than purely geometric distance. This TTC-aware strategy introduces velocity-dependent obstacle avoidance behavior while preserving the lightweight reactive structure characteristic of FTG approaches.

All reactive methods operate under the same AEBS supervisory layer, enabling direct comparison between purely reactive obstacle avoidance and safety-supervised navigation behaviors.

III. EXPERIMENTAL SETUP

The evaluated navigation methods were tested under identical simulation conditions using the unified ROS2 and Gazebo framework described previously. All experiments were conducted on a differential-drive robot navigating a static obstacle-constrained environment represented through occupancy maps and collision meshes. The simulated map covered an area of approximately $42\text{ m} \times 22\text{ m}$. For map-based navigation, the same initial pose, goal position, and ARA*-generated reference trajectory were used across all trials to ensure fair comparison between tracking controllers.

The robotic platform was equipped with a 360° LiDAR sensor providing 720 range measurements per scan with an effective sensing range between 0.1 m and 12 m. All navigation controllers operated at a control frequency of 25 Hz. The evaluated methods include two map-based controllers (Pure Pursuit and Stanley) and three reactive navigation strategies (wall-following, distance-based Follow-the-Gap, and TTC-based Follow-the-Gap). All controllers were manually tuned to achieve stable navigation behavior while maintaining comparable operating conditions and velocity ranges. Additionally, all methods operated under the same TTC-based AEBS supervisory layer.

All experiments started from the same initial pose with zero linear and angular velocity. Since the evaluated simulations were deterministic under fixed initial conditions and static environments, representative single-run traversals were used for quantitative analysis. Repeated executions under identical conditions exhibited consistent qualitative behavior and comparable performance trends across all evaluated methods.

Performance evaluation focused on navigation efficiency, trajectory quality, motion smoothness, and safety-related behavior. The analyzed metrics include traversal completion time, path length, root-mean-square (RMSE) tracking error, mean and standard deviation of linear and angular velocities, and the number of AEBS braking activations. For map-based methods, tracking error was computed as the Euclidean distance between the executed trajectory and the closest point on the reference path at each time step.

IV. RESULTS AND ANALYSIS

All simulations were executed on a workstation equipped with an Intel(R) Core(TM) i7-10870H CPU @ 2.20 GHz (8 cores), 16 GB of RAM, an NVIDIA GeForce GTX 1650 Ti GPU, and solid-state storage. The operating system was Ubuntu 24.04 LTS running ROS2 Jazzy and Gazebo Harmonic. All navigation methods were implemented as distributed ROS2 nodes and executed under identical computational conditions to ensure fair comparison and reproducibility.

The source code, recorded simulation videos, and numerical results corresponding to all evaluated methods are provided in the Supplementary Material, enabling reproducibility and qualitative inspection of the navigation behavior.

Table I summarizes the principal quantitative metrics obtained for all evaluated navigation methods.

The map-based controllers exhibited substantially different trajectory-tracking behaviors despite operating over the same precomputed path. Pure Pursuit achieved the highest mean linear velocity together with relatively low velocity variability, indicating smoother forward motion and more stable trajectory execution. In contrast, Stanley presented significantly larger variability in both linear and angular velocities, reflecting more aggressive corrective steering actions and less consistent speed regulation.

Both controllers exhibit negative mean angular velocity because the evaluated trajectory is traversed predominantly in the clockwise direction. However, Stanley produced considerably higher angular velocity variability, indicating frequent and abrupt steering corrections consistent with its nonlinear cross-track feedback formulation. These behaviors were particularly evident in regions containing consecutive curvature changes, where the controller repeatedly overcorrected before previous deviations fully settled.

From a tracking perspective, Pure Pursuit achieved lower mean tracking error and lower RMSE, demonstrating closer adherence to the reference trajectory. Although both controllers experienced localized deviations in constrained or high-curvature regions, the Pure Pursuit trajectory remained smooth and bounded throughout the traversal. In contrast, Stanley exhibited oscillatory behavior resembling an under-damped response, producing sharper heading corrections and locally jagged trajectories.

Fig. 1 shows the executed trajectory of the Pure Pursuit controller overlaid on the planned path. The controller follows the global structure smoothly, with gradual curvature transitions and no visible oscillatory behavior. Minor corner-cutting effects are observed in high-curvature regions due to the geometric look-ahead formulation, though these deviations remain localized and preserve obstacle clearance margins.

Fig. 2 presents the Stanley trajectory under identical conditions. While the controller maintains close adherence along straight segments, oscillatory corrections become visible in regions with frequent heading changes. Several trajectory sections exhibit overshoot and repeated corrective actions, con-

¹Tracking-error metrics are omitted for reactive methods because these approaches do not follow a predefined global reference trajectory.

TABLE I
UNIFIED NAVIGATION PERFORMANCE METRICS¹

Method	Linear Velocity [m/s]		Angular Velocity [rad/s]		Tracking Error [m]		AEBS	Path Length [m]	Completion Time [s]
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	RMSE			
Pure Pursuit	3.866	0.577	-0.244	0.836	0.191	0.297	3	103.384	27.840
Stanley	2.034	1.150	-0.123	1.033	0.272	0.367	16	110.513	57.099
Wall-Following	2.375	0.580	-0.170	0.817	–	–	11	105.346	48.699
FTG (Distance)	1.165	0.513	-0.070	0.492	–	–	17	103.092	104.799
FTG (TTC)	2.094	0.230	-0.140	0.432	–	–	0	94.128	45.249

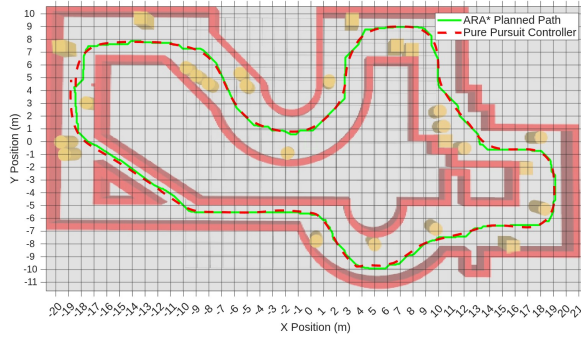


Fig. 1. Executed trajectory of the Pure Pursuit controller overlaid on the planned reference path.

tributing to increased tracking error, reduced motion smoothness, and temporary reductions in obstacle clearance.

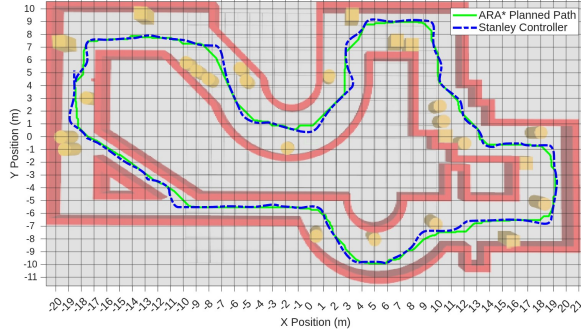


Fig. 2. Executed trajectory of the Stanley controller overlaid on the planned reference path.

The reactive navigation methods instead emphasized local obstacle negotiation rather than strict path adherence. Among these approaches, wall-following achieved relatively high forward velocity but also exhibited the largest motion variability, indicating stronger local steering corrections near boundaries and obstacle regions. The distance-based Follow-the-Gap (FTG) method produced the lowest mean velocity and the longest traversal time, reflecting a conservative obstacle-avoidance behavior during narrow-gap negotiation. In contrast, the TTC-based FTG maintained comparatively high forward velocity together with low velocity variability, indicating smoother and more anticipatory obstacle avoidance behavior.

The influence of the TTC-based Automatic Emergency Braking System is particularly evident in the safety metrics. Stanley and distance-based FTG produced the largest number of AEBS activations, indicating that both methods more fre-

quently operated near collision-risk conditions. For Stanley, these interventions were primarily associated with oscillatory corrections in constrained regions, whereas for distance-based FTG they resulted from late geometric gap selection near nearby obstacles. Conversely, TTC-based FTG completed the traversal without a single AEBS activation. Since obstacle selection in this method explicitly incorporates collision-time estimation, the controller tends to initiate obstacle avoidance maneuvers earlier and avoids trajectories likely to violate the TTC safety threshold.

Efficiency metrics further differentiate the evaluated methods. Pure Pursuit achieved the shortest completion time among all evaluated approaches while simultaneously maintaining relatively low tracking error and reduced safety intervention frequency. Stanley required substantially longer traversal time due to repeated corrective maneuvers and emergency braking events. Among the reactive methods, TTC-based FTG achieved significantly faster traversal than the distance-based formulation while maintaining smoother behavior and safer obstacle negotiation. Although wall-following preserved relatively high average velocity, its stronger local corrections and frequent AEBS interventions reduced overall efficiency in obstacle-rich regions.

Fig. 3 presents representative trajectories of the reactive navigation strategies in the obstacle-rich environment. The wall-following method remains attached to nearby boundaries and successfully negotiates inserted obstacles, though with sharper local corrections consistent with its increased angular variability. Distance-based FTG selects narrower gaps and exhibits visibly more conservative progression, particularly near obstacle clusters where repeated AEBS activations occur. In contrast, TTC-based FTG produces smoother curvature transitions and earlier obstacle avoidance maneuvers, preserving trajectory continuity while avoiding emergency braking interventions entirely.

The results highlight clear trade-offs among tracking aggressiveness, motion smoothness, obstacle anticipation, and safety-supervised navigation performance. Controllers relying on smoother geometric convergence or TTC-aware obstacle selection generally produced fewer safety interventions and more efficient traversal behavior. In contrast, methods characterized by stronger corrective feedback or purely geometric obstacle selection exhibited increased oscillatory motion, reduced traversal efficiency, and more frequent operation near collision-risk conditions. Under the evaluated conditions, Pure Pursuit and TTC-based FTG demonstrated the most consistent balance among tracking stability, navigation smoothness,

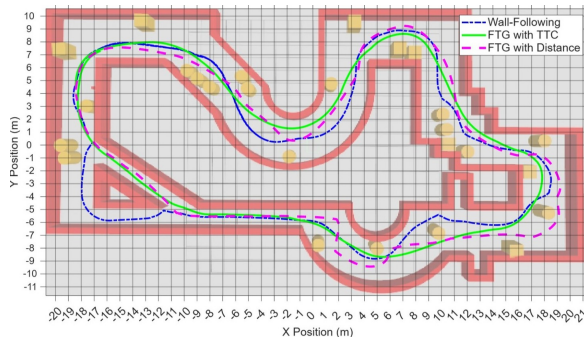


Fig. 3. Representative trajectories of the reactive navigation methods in the obstacle-rich environment.

traversal efficiency, and safety-aware operation.

V. CONCLUSION

This work presented a unified simulation-based evaluation framework for autonomous navigation of a differential-drive robot integrating both reactive and map-based navigation strategies within a common ROS2 and Gazebo architecture. The evaluated methods included Pure Pursuit and Stanley path-tracking controllers, together with wall-following, distance-based Follow-the-Gap (FTG), and TTC-based FTG reactive approaches, all operating under the same TTC-based Automatic Emergency Braking System (AEBS) supervisory layer.

The results revealed clear behavioral differences among the evaluated navigation paradigms. Among the map-based methods, Pure Pursuit achieved smoother trajectory execution, lower tracking error, fewer safety interventions, and shorter traversal time than Stanley, whose nonlinear corrective behavior produced oscillations and increased AEBS activations in constrained regions. Within the reactive approaches, the TTC-based FTG formulation exhibited the most consistent behavior, maintaining relatively high forward velocity while avoiding emergency braking events through anticipatory obstacle avoidance. In contrast, wall-following and distance-based FTG generated more abrupt local corrections and higher safety-intervention frequency, particularly in obstacle-rich environments.

The study highlights the importance of balancing tracking aggressiveness, motion smoothness, and anticipatory collision handling in autonomous navigation systems. The results additionally suggest that incorporating time-to-collision reasoning directly into navigation decisions can improve safety-related behavior while preserving traversal efficiency.

The proposed framework is limited to static simulated environments and manually tuned controller parameters, and the observed behaviors may vary under different geometries or operating conditions. Future work will extend the framework to dynamic environments, multi-run statistical evaluation, and

additional navigation strategies, including learning-based and optimization-based approaches. Experimental validation on physical robotic platforms is also planned to assess the practical transferability of the reported behaviors.

SUPPLEMENTARY MATERIAL

The simulation videos and the source code used in this work, along side the numerical results, are available online:

- Reactive Methods Simulation videos: <https://youtube.com/playlist?list=PLPTNNUtNHvbhvaN7qWtmGJkLKGyXz8kOK&si=IcljmTEIdLOKOUax>
- Reactive Methods GitHub repository: https://github.com/santy-estrada/mobile_robotics_class
- Map-Based Methods Simulation videos: <https://www.youtube.com/playlist?list=PLPTNNUtNHvbhvaB116Peh17-pD8-NkMfL>
- Map-Based Methods GitHub repository: https://github.com/santy-estrada/map_based_methods_diffdrive

REFERENCES

- [1] D. Fox, W. Burgard, and S. Thrun, "The dynamic window approach to collision avoidance," *IEEE Robotics & Automation Magazine*, vol. 4, no. 1, pp. 23–33, 1997.
- [2] E. Baklouti, N. B. Amor, and M. Jallouli, "Reactive control architecture for mobile robot autonomous navigation," *Robotics and Autonomous Systems*, vol. 89, pp. 9–14, 2017.
- [3] V. Sezer and M. Gokasan, "A novel obstacle avoidance algorithm: "follow the gap method", " *Robotics and Autonomous Systems*, vol. 60, no. 9, pp. 1123–1134, 2012.
- [4] J. Toibero, R. Carelli, and B. Kuchen, "Wall-following stable control for wheeled mobile robots," *IFAC Proceedings Volumes*, vol. 39, no. 15, pp. 85–90, 2006.
- [5] V. Chernov and A. S. Matveev, "Reactive navigation of nonholonomic robots for search and tight circumnavigation of group objects through singular inter-object gaps," *Robotics and Autonomous Systems*, vol. 174, p. 104649, 2024.
- [6] L. Jianqing, L. Zilong, X. Jingjie, and S. Zhaohui, "Dynamic path planning for multiple unmanned ground vehicles based on improved dynamic window approach," *Chinese Journal of Aeronautics*, p. 104007, 2025.
- [7] J. M. Snider, "Automatic steering methods for autonomous automobile path tracking," Technical Report CMU-RI-TR-09-08, Robotics Institute, Carnegie Mellon University, Pittsburgh, PA, USA, 2009.
- [8] S. Thrun, M. Montemerlo, H. Dahlkamp, D. Stavens, A. Aron, J. Diebel, P. Fong, J. Gale, M. Halpenny, G. Hoffmann, *et al.*, "Stanley: The robot that won the darpa grand challenge," *Journal of Field Robotics*, vol. 23, pp. 661–692, Sept. 2006.
- [9] M. Samuel, M. H. Hussein, and M. B. Mohamad, "A review of some pure-pursuit based path tracking techniques for control of autonomous vehicle," *International Journal of Computer Applications*, vol. 135, pp. 35–38, Feb. 2016.
- [10] M. Likhachev, G. Gordon, and S. Thrun, "Ara*: Anytime a* with provable bounds on sub-optimality," in *Proceedings of the 17th International Conference on Neural Information Processing Systems (NIPS)*, pp. 767–774, 2003.
- [11] A. Jafari and Y.-C. Liu, "Pedestrians' safety using projected time-to-collision to electric scooters," *Nature Communications*, vol. 15, no. 1, p. 5701, 2024.